

MK-UNet on a 6GB GPU

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Two independently trained models were evaluated: the official MK-UNet architecture on the authors' ClinicDB split, and the same architecture trained from scratch on Massachusetts Buildings. All values below come from saved checkpoints and actual local runs. This is a single-seed reproduction attempt, not an exact reproduction of the paper's five-run average.

Measure	ClinicDB reference experiment	Massachusetts Buildings
Training / validation / test	489 / 61 / 62 images	137 / 4 / 10 complete tiles
Input	352 x 352; scales 256, 352, 448	1500 x 1500 tiles resized to 512 x 512; four 256 x 256 patches
Optimizer / epochs	AdamW; lr = 1e-4; wd = 1e-4; 200 epochs	Same optimizer; 200 epochs; single-scale
Precision / augmentation	Mixed precision; no augmentation	FP32; no augmentation
Selected epoch	58	136
Native mean Dice, raw sigmoid	89.59%	71.95%
Native mean IoU, raw sigmoid	83.62%	56.25%

Environment and evidence

NVIDIA GeForce RTX 4050 Laptop GPU (6GB); Windows; PyTorch 2.6.0+cu124; CUDA 12.4; timm 0.9.16. The official architecture is unmodified and contains 315,566 parameters. Source revision and file SHA-256 hashes are recorded in each run's configuration and data manifest.

ClinicDB elapsed run time: 67.4 minutes; reported peak tensor allocation: 5095 MiB. Massachusetts: 15.8 minutes; 3282 MiB. Timing includes validation and test-data loading, but excludes the final test-scoring passes. Allocation includes cached dataset tensors and is not an architecture-only memory estimate.

Working code and recorded runs

How to reproduce and audit

Commands and preserved artifacts

Run `python download_data.py clinic` and `python download_data.py mass` from the archive's project root. Then run `python train_screening.py --dataset clinic --epochs 200 --batch 16 --amp` and `python train_screening.py --dataset mass --epochs 200 --batch 16`. See README for environment setup and directory conventions. Dataset downloads are excluded from the archive.

For each experiment: `config.json` records the environment and decisions; `history.jsonl` records every epoch; `best.pt` and `last.pt` preserve checkpoints; `data_manifest.json` records sample identities and hashes; `results.json` contains native-resolution aggregate and per-image results. Raw console logs accompany the epoch logs. The pilot directory records stress conditions, per-case adaptation decisions and metrics.

Protocol choices and codebase observations

The paper uses AdamW at $1e-4$, batch size 16, 200 epochs and no augmentation. The current repository script instead defaults to lr $5e-4$ with a cosine scheduler, batch size 8 and augmentation enabled. The wrapper follows the stated paper settings and omits the scheduler. Mixed precision is a declared hardware-driven change. The model is seeded, but cuDNN benchmarking is enabled; bitwise repeatability is not claimed.

Input resizing uses PIL rather than upstream OpenCV/Albumentations, and masks use the fixed >127 rule rather than upstream >20 . The official loader returns PIL image dimensions as (width, height), while the evaluator names them (height, width). The wrapper obtains the native mask shape directly, preserves its labels, resizes logits to that shape, and thresholds the original mask at 127. It also resets each multiscale input from the original batch; the current script reassigns images within the scale loop. Validation alone selects the checkpoint, and the held-out test metrics are computed after training.

Metric definitions and comparison limits

$\text{Dice} = 2\text{TP} / (2\text{TP} + \text{FP} + \text{FN})$; $\text{IoU} = \text{TP} / (\text{TP} + \text{FP} + \text{FN})$, with a $1e-6$ stabilizer. Primary metrics average per original image or tile, after bilinear logit resizing and sigmoid ≥ 0.5 . An all-empty prediction and all-empty target score 1. No test labels are used for threshold tuning. Massachusetts logits are stitched into a 512×512 tile before resizing to its original 1500×1500 mask.

For comparison to the code's normalization convention, native ClinicDB Dice after per-image min-max normalization is 89.59%. The paper reports 93.48% averaged over five runs [1]; our signed difference is -3.89 percentage points. Differences in precision, software, evaluator corrections and run count prevent an exact-equivalence claim. The authors' frame split may share video sequences; exact-image hash separation does not establish patient or sequence separation.

Massachusetts is a supervised transfer of the architecture to a different task. Training a new model on its labels is not evidence that a ClinicDB model generalizes zero-shot to buildings. The official tile split prevents patches from the same tile crossing partitions; neighboring tiles may still be spatially correlated.

Sources: [1], [2], [11], [12] in the proposal bibliography. Working code and recorded runs